THE BOUNDED-ABOVE DERIVED FUNCTOR OF THE SERRE QUOTIENT EMBEDDING IS AN EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6370-6416, following the statement at 6360-6368.

Editorial note. 'Fully faithfulness is proved in the exact same manner' (6414-6415): mirror of the executed faithfulness argument with the homotopy applied to the other composite, recorded as a routine repeat.

Allowed standard prerequisites (editorial).

- Serre subcategory closure. the Serre hypothesis of the statement, used at proof line 6399. Supplies that (D^{i+1} + E^{i+1}) intersect Im(d^i) lies in B, making the inductive choice of E^i possible.

Source reference. Original equation 06UR: sources/stacks-derived/pinned-original-source.tex, source lines 6188–6188. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{B} \subset \mathcal{A}$ be a Serre subcategory. Assume that for every surjection $X \to Y$ with $X \in \mathrm{Ob}(\mathcal{A})$ and $Y \in \mathrm{Ob}(\mathcal{B})$ there exists $X' \subset X$, $X' \in \mathrm{Ob}(\mathcal{B})$ which surjects onto Y. Then the functor $D^-(\mathcal{B}) \to D^-_{\mathcal{B}}(\mathcal{A})$ of (06UR) is an equivalence.*

Proof. Let $X^\bullet$ be a bounded above complex of $\mathcal{A}$ such that $H^i(X^\bullet) \in \mathrm{Ob}(\mathcal{B})$ for all $i \in \mathbf{Z}$. Moreover, suppose we are given $B^i \subset X^i$, $B^i \in \mathrm{Ob}(\mathcal{B})$ for all $i \in \mathbf{Z}$. Claim: there exists a subcomplex $Y^\bullet \subset X^\bullet$ such that

(1) $Y^\bullet \to X^\bullet$ is a quasi-isomorphism,
(2) $Y^i \in \mathrm{Ob}(\mathcal{B})$ for all $i \in \mathbf{Z}$, and
(3) $B^i \subset Y^i$ for all $i \in \mathbf{Z}$.

To prove the claim, using the assumption of the lemma we first choose $C^i \subset \mathrm{Ker}(d^i : X^i \to X^{i+1})$, $C^i \in \mathrm{Ob}(\mathcal{B})$ surjecting onto $H^i(X^\bullet)$. Setting $D^i = C^i + d^{i-1}(B^{i-1}) + B^i$ we find a subcomplex $D^\bullet$ satisfying (2) and (3) such that $H^i(D^\bullet) \to H^i(X^\bullet)$ is surjective for all $i \in \mathbf{Z}$. For any choice of $E^i \subset X^i$ with $E^i \in \mathrm{Ob}(\mathcal{B})$ and $d^i(E^i) \subset D^{i+1} + E^{i+1}$ we see that setting $Y^i = D^i + E^i$ gives a subcomplex whose terms are in $\mathcal{B}$ and whose cohomology surjects onto the cohomology of $X^\bullet$. Clearly, if $d^i(E^i) = (D^{i+1} + E^{i+1}) \cap \mathrm{Im}(d^i)$ then we see that the map on cohomology is also injective. For $n \gg 0$ we can take E^n equal to 0. By descending induction we can choose E^i for all i with the desired property. Namely, given $E^{i+1}, E^{i+2}, \ldots$ we choose $E^i \subset X^i$ such that $d^i(E^i) = (D^{i+1} + E^{i+1}) \cap \mathrm{Im}(d^i)$. This is possible by our

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assumption in the lemma combined with the fact that $(D^{i+1} + E^{i+1}) \cap \mathrm{Im}(d^i)$ is in $\mathcal{B}$ as $\mathcal{B}$ is a Serre subcategory of $\mathcal{A}$.

The claim above implies the lemma. Essential surjectivity is immediate from the claim. Let us prove faithfulness. Namely, suppose we have a morphism $f : U^\bullet \to V^\bullet$ of bounded above complexes of $\mathcal{B}$ whose image in $D(\mathcal{A})$ is zero. Then there exists a quasi-isomorphism $s : V^\bullet \to X^\bullet$ into a bounded above complex of $\mathcal{A}$ such that $s \circ f$ is homotopic to zero. Choose a homotopy $h^i : U^i \to X^{i-1}$ between 0 and $s \circ f$. Apply the claim with $B^i = h^{i+1}(U^{i+1}) + s^i(V^i)$. The resulting map $s' : V^\bullet \to Y^\bullet$ is a quasi-isomorphism as well and $s' \circ f$ is homotopic to zero as is clear from the fact that h^i factors through Y^{i-1}. This proves faithfulness. Fully faithfulness is proved in the exact same manner. $\square$